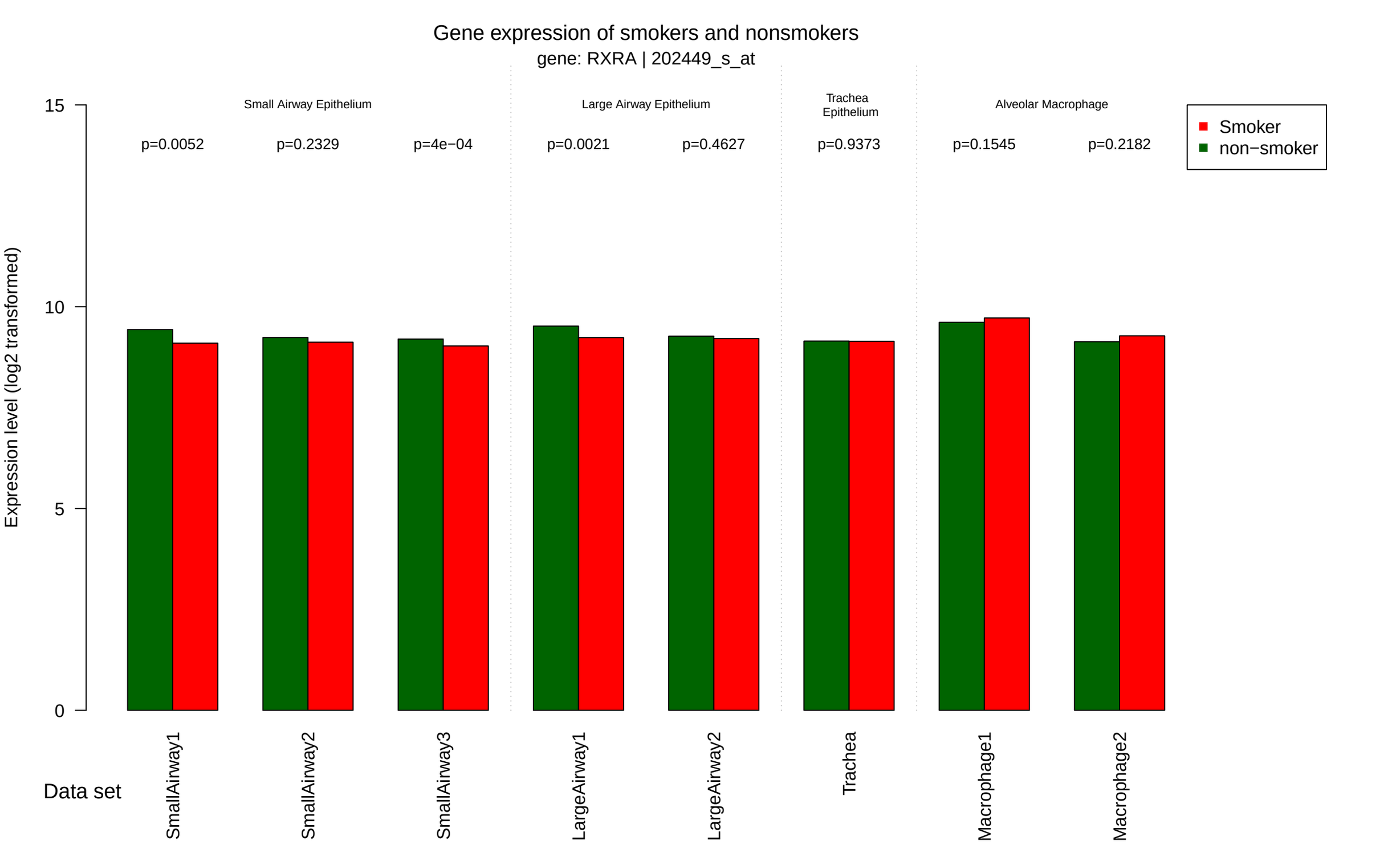
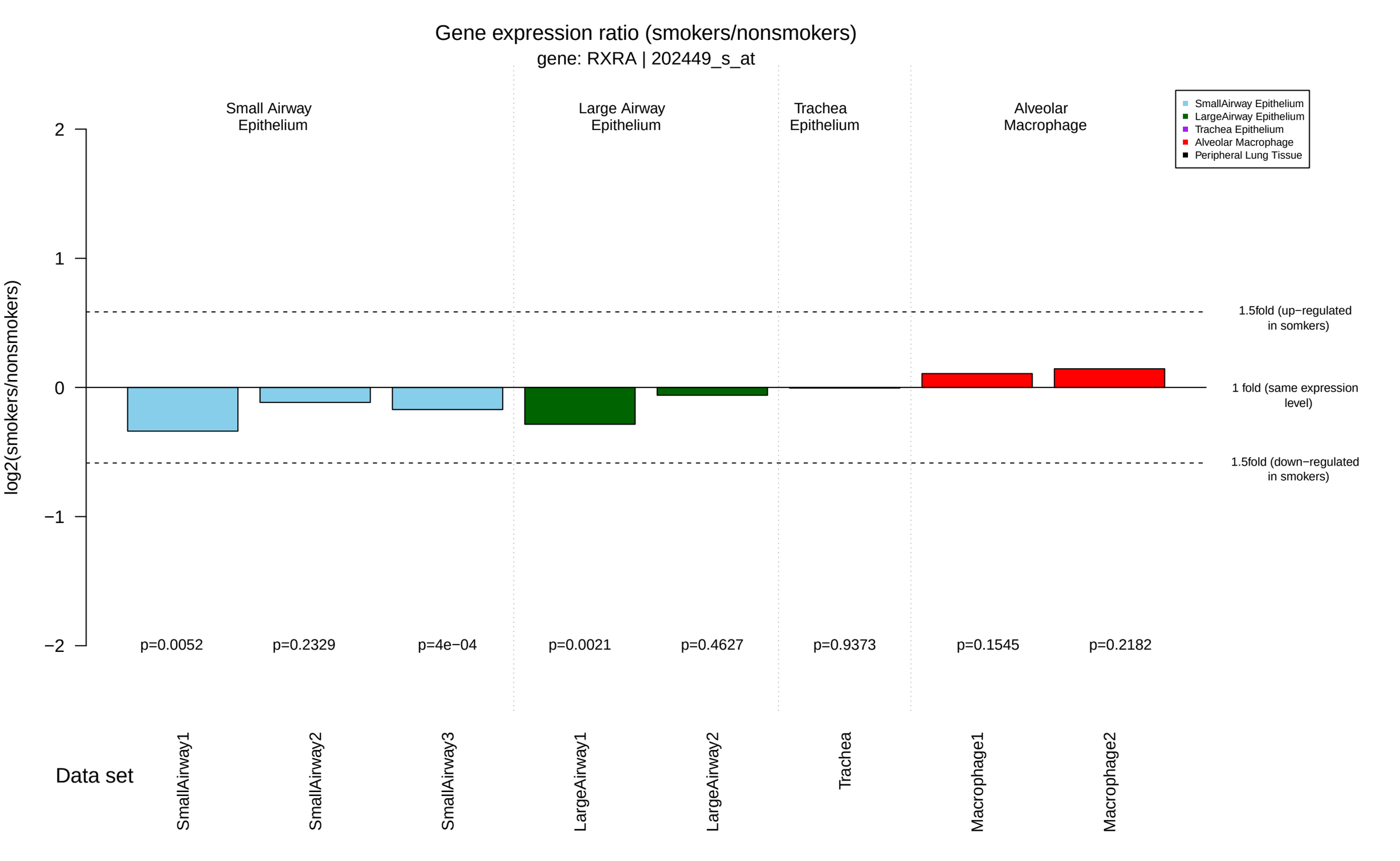
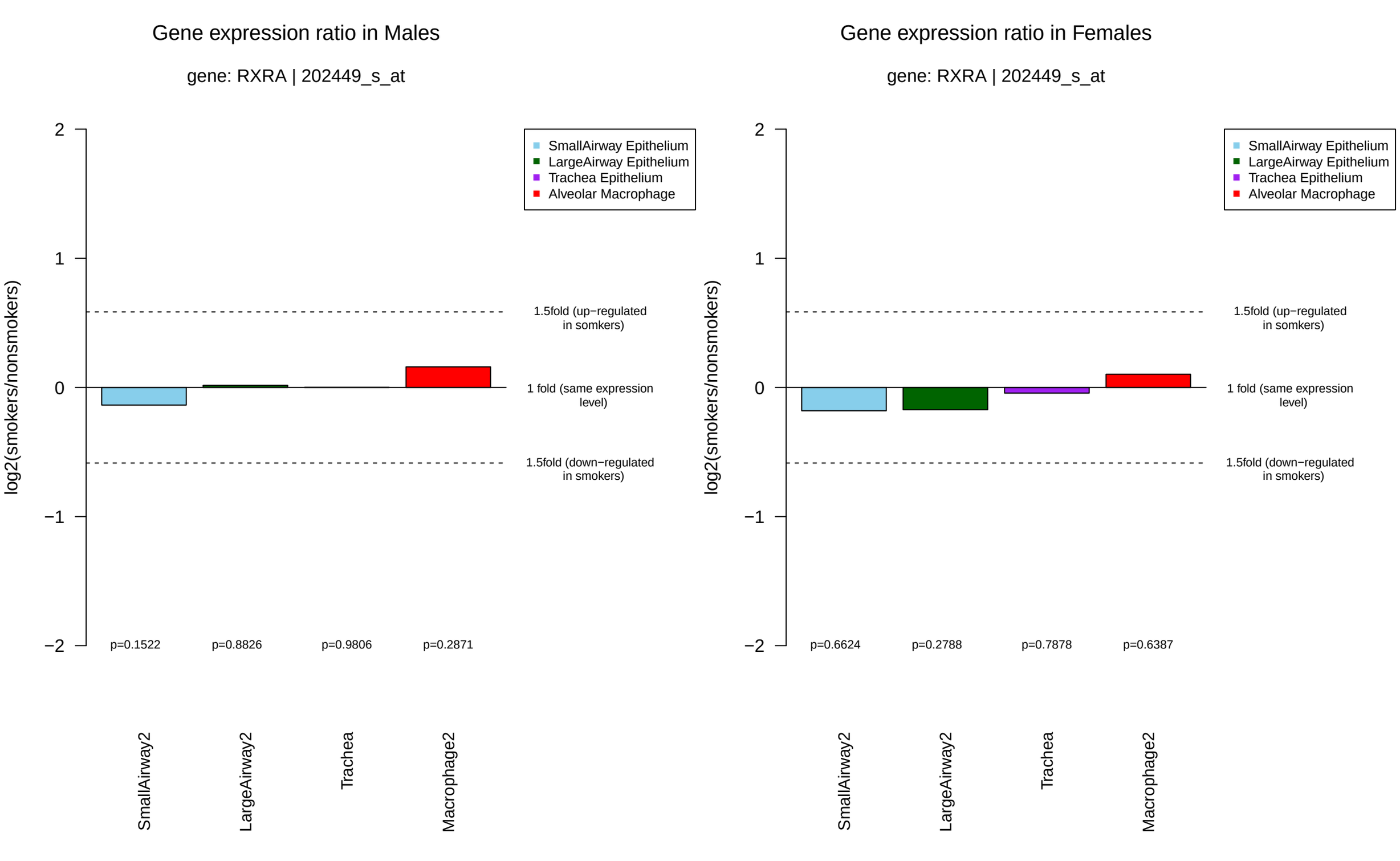
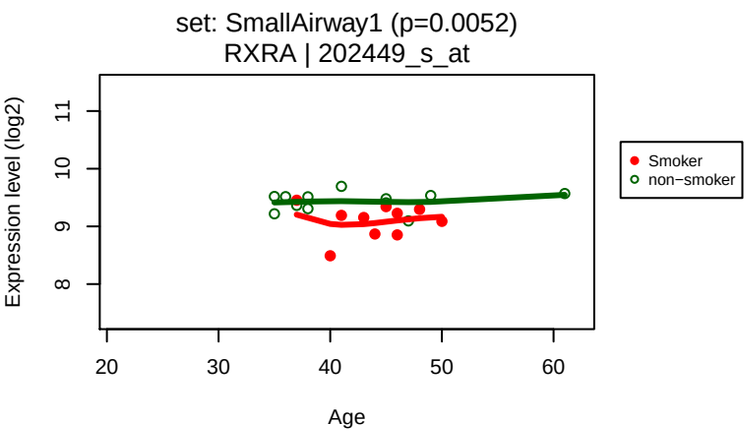
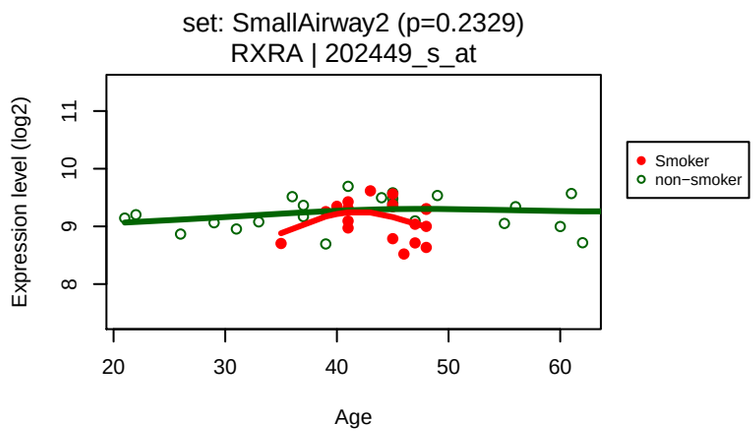
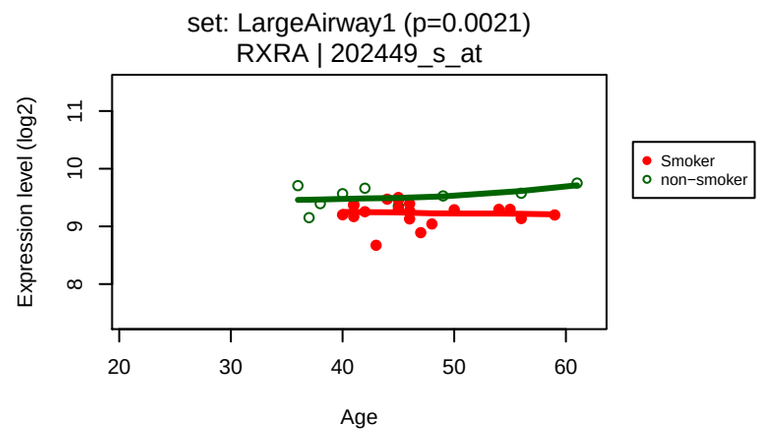

Gene expression of smokers and nonsmokers
gene: RXRA | 202449_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0052
p=0.2329
p=4e−04
p=0.0021
p=0.4627
p=0.9373
p=0.1545
p=0.2182
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

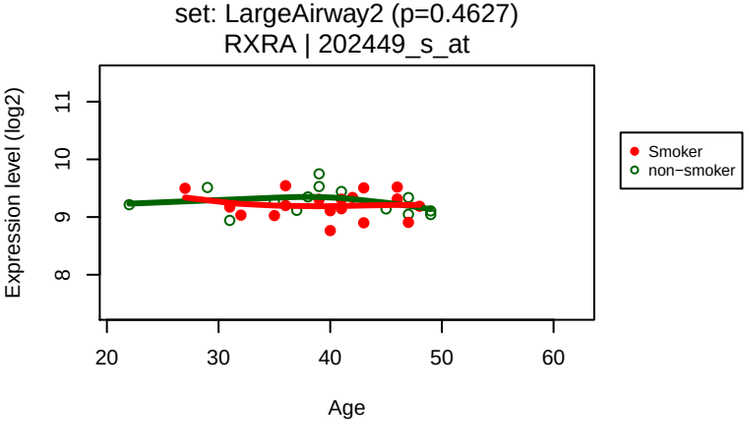

Gene expression ratio (smokers/nonsmokers)
gene: RXRA | 202449_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0052
p=0.2329
p=4e−04
p=0.0021
p=0.4627
p=0.9373
p=0.1545
p=0.2182
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

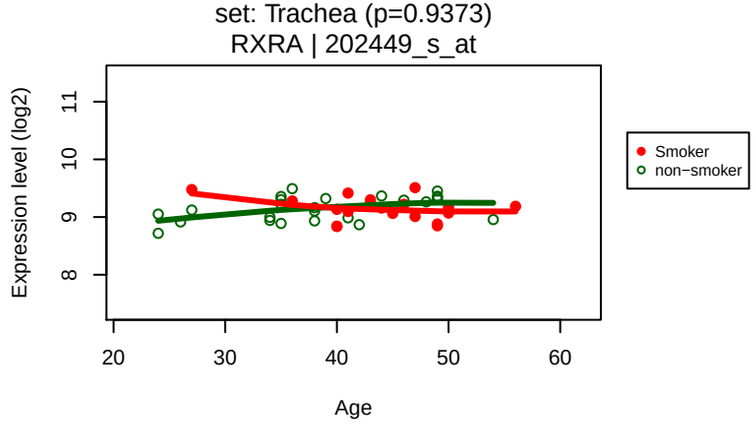

Gene expression ratio in Males
gene: RXRA | 202449_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1522
p=0.8826
p=0.9806
p=0.2871
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RXRA | 202449_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6624
p=0.2788
p=0.7878
p=0.6387
SmallAirway2
LargeAirway2
Trachea
Macrophage2

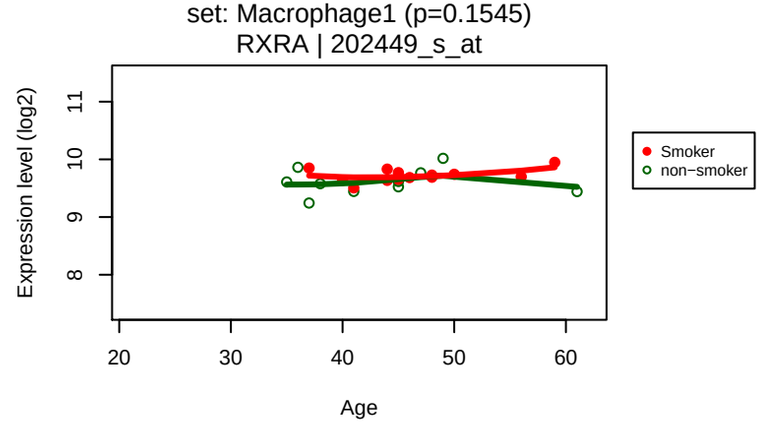

set: SmallAirway1 (p=0.0052)
RXRA | 202449_s_at
Expression level (log2)
Age
Smoker
non−smoker

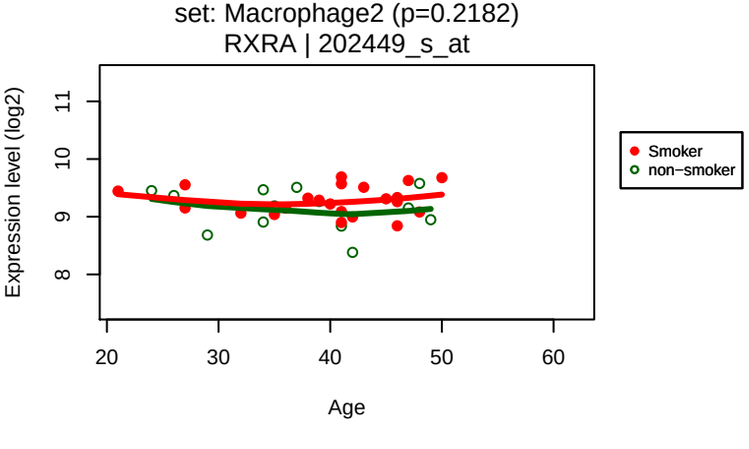

set: SmallAirway2 (p=0.2329)
RXRA | 202449_s_at
Expression level (log2)
Age
Smoker
non−smoker

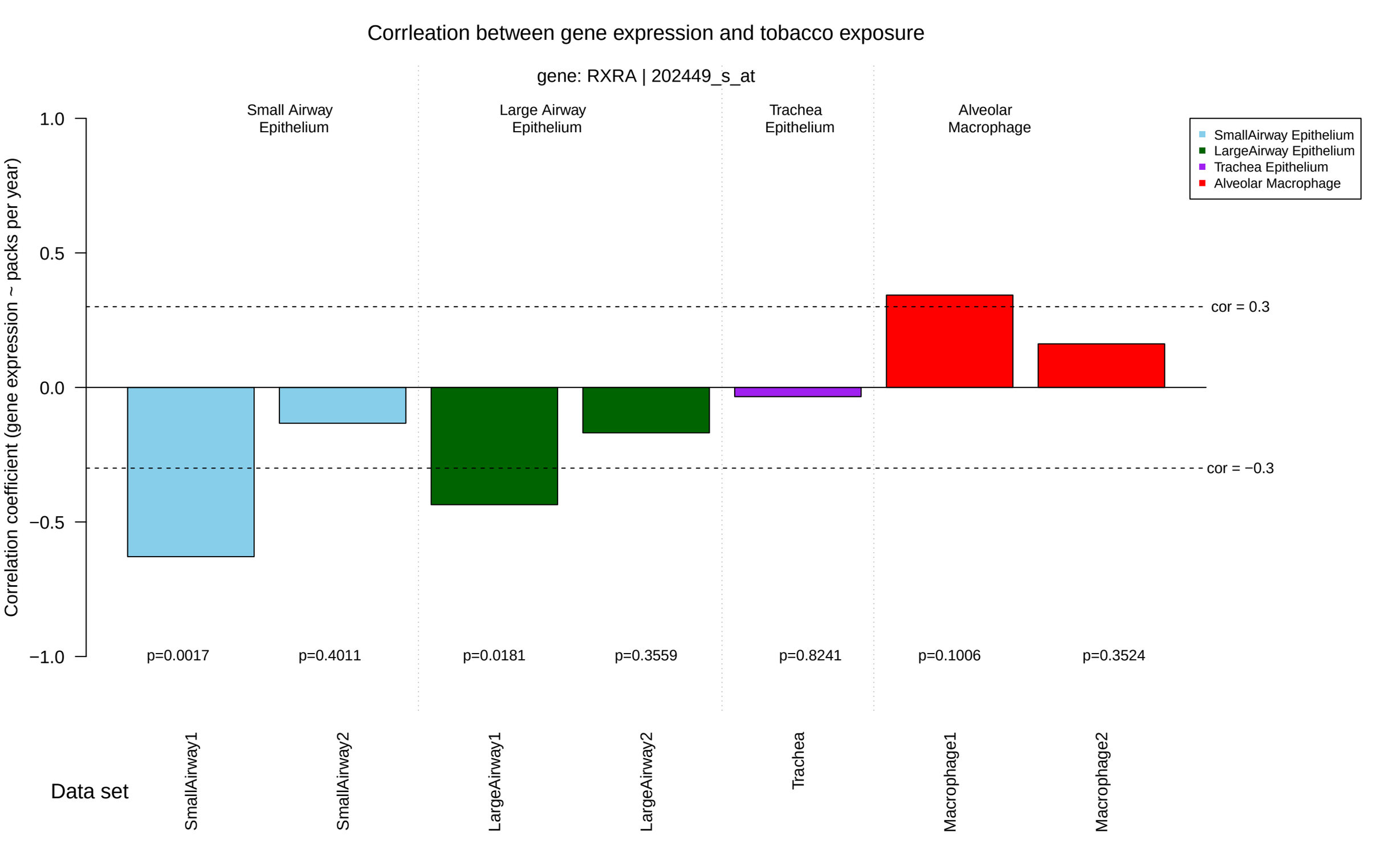

set: LargeAirway1 (p=0.0021)
RXRA | 202449_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4627)
RXRA | 202449_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9373)
RXRA | 202449_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1545)
RXRA | 202449_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2182)
RXRA | 202449_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RXRA | 202449_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0017
p=0.4011
p=0.0181
p=0.3559
p=0.8241
p=0.1006
p=0.3524
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2